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Machine learning to predict the long-term risk of myocardial infarction and cardiac death based on clinical risk, coronary calcium, and epicardial adipose tissue: a prospective study

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Aims Our aim was to evaluate the performance of machine learning (ML), integrating clinical parameters with coronary artery calcium (CAC), and automated epicardial adipose tissue (EAT) quantification, for the prediction of long-term risk of myocardial infarction (MI) and cardiac death in asymptomatic subjects.

Methods and results Our study included 1912 asymptomatic subjects [1117 (58.4%) male, age: 55.8 ± 9.1 years] from the prospective EISNER trial with long-term follow-up after CAC scoring. EAT volume and density were quantified using a fully automated deep learning method. ML extreme gradient boosting was trained using clinical co-variates, plasma lipid panel measurements, risk factors, CAC, aortic calcium, and automated EAT measures, and validated using repeated 10-fold cross validation. During mean follow-up of 14.5 ± 2 years, 76 events of MI and/or cardiac death occurred. ML obtained a significantly higher AUC than atherosclerotic cardiovascular disease (ASCVD) risk and CAC score for predicting events (ML: 0.82; ASCVD: 0.77; CAC: 0.77, $P < 0.05$ for all). Subjects with a higher ML score (by Youden's index) had high hazard of suffering events (HR: 10.38, $P < 0.001$); the relationships persisted in multivariable analysis including ASCVD-risk and CAC measures (HR: 2.94, $P = 0.005$). Age, ASCVD-risk, and CAC were prognostically important for both genders. Systolic blood pressure was more important than cholesterol in women, and the opposite in men.

Conclusions In this prospective study, machine learning used to integrate clinical and quantitative imaging-based variables significantly improves prediction of MI and cardiac death compared with standard clinical risk assessment. Following further validation, such a personalized paradigm could potentially be used to improve cardiovascular risk assessment.

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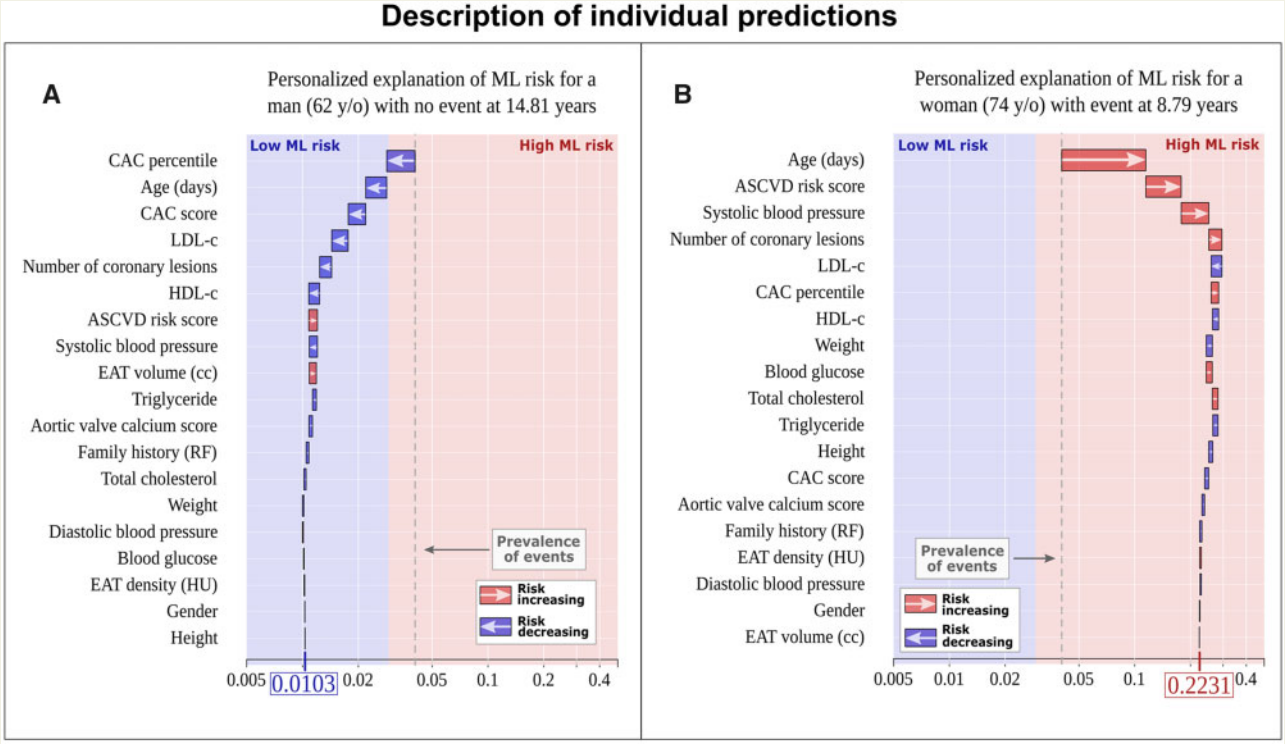
Graphical Abstract



Keywords Machine learning • Epicardial adipose tissue • Coronary calcium scoring • Myocardial infarction and cardiac death

1. Introduction

Coronary artery calcium (CAC) scoring with non-contrast computed tomography (CT) is increasingly used for cardiovascular risk stratification and provides a measure of coronary atherosclerotic burden. To date, several studies have reported that the total CAC measured by non-contrast CT predicts cardiovascular events,[1–15] beyond standard cardiovascular risk factors. The prospective, randomized EISNER (Early Identification of Subclinical Atherosclerosis by Noninvasive Imaging Research) trial[16] has reported that CAC scoring improves control of cardiovascular risk factors in asymptomatic patients and is not associated with increased downstream medical testing or costs.[16,17] Beyond CAC scoring, epicardial adipose tissue (EAT), a metabolically active fat deposit, has been shown to relate to early atherosclerosis[18–21] and to predict adverse cardiovascular events.[22] However, machine learning (ML) risk scores—which efficiently maximize the information from all available clinical risk factors, as well as CAC, aortic calcium, and EAT measurements from low-cost, low-radiation dose non-contrast CT scans— are not yet utilized after CAC scanning for the prediction of future cardiovascular events.

ML techniques allow for objective integration of data to predict adverse outcomes. ML has been used to predict all-cause mortality in patients with suspected coronary artery disease (CAD) undergoing coronary CT angiography by integrating clinical and visually assessed CT angiography parameters.[23] However, the use of ML to perform global cardiovascular risk stratification in asymptomatic patients from clinical, CAC, and EAT quantification has not yet been evaluated.

In this study, we aimed to evaluate a boosting approach to ML, which has shown significant higher performance than other ML techniques, for objective prediction of myocardial infarction (MI) and cardiac death in asymptomatic subjects from the CT imaging arm of the EISNER trial. We also evaluated ML for men and women separately to determine sex-based differences in the prognostic importance of covariates.

2. Methods

2.1 Study population

Our study consisted of consecutive asymptomatic subjects from the prospective EISNER trial who underwent CAC scoring with available cardiac CT scans and long-term follow-up (965 patients from the CAC-scan group from the randomized EISNER trial,[16] and 947 subjects with available cardiac CT scans and long-term follow-up from the EISNER 4 sub-trial with no randomization). As described previously, the EISNER trial represents a community-based cohort of asymptomatic subjects who underwent complete cardiovascular risk assessment.[16] The EISNER participants were middle-aged subjects, with cardiovascular risk factors, but no known CAD. Subjects with any cardiac or cerebrovascular

Table 1 Population and cardiovascular risk factors

Characteristics	Total subjects	Event	No event	*P*-value
Age (years)	55.8 ± 9.1	62.1 ± 11.7	55.6 ± 8.8	**<0.001**
Gender (male)	1117 (58.4%)	51 (67.1%)	1066 (58.1%)	0.117
Weight (lb)	170 (145.0–192.0)	175 (155.0–197.8)	169 (144.0–192.0)	0.062
Systolic blood pressure (mmHg)	127 (115.0–140.0)	138 (128.8–151.3)	127 (115.0–140.0)	**<0.001**
Diastolic blood pressure (mmHg)	78 (69.0–85.0)	80.5 (72.8–89.3)	77 (69.0–84.0)	**0.002**
LDL-cholesterol (mg/dL)	129 (106.0–152.0)	142 (116.3–166.3)	129 (106.0–151.0)	**0.002**
HDL-cholesterol (mg/dL)	53 (43.0–65.5)	46 (38.0–59.0)	53 (43.0–66.0)	**<0.001**
Total cholesterol (mg/dL)	210 (184.0–235.0)	221 (195.8–239.0)	209 (183.3–234.8)	0.053
Triglycerides (mg/dL)	104 (73.0–151.0)	124 (92.5–161.5)	103 (72.0–151.0)	**0.006**
Cardiovascular risk factors				
Hypertension	744 (38.9%)	47 (61.8%)	697 (38.0%)	**<0.001**
Hypercholesterolemia	1318 (68.9%)	54 (71.1%)	1264 (68.8%)	0.684
Diabetes mellitus	104 (5.4%)	5 (6.6%)	99 (5.4%)	0.655
Smoking	120 (6.3%)	6 (7.9%)	114 (6.2%)	0.553
Family history	570 (29.8%)	27 (35.5%)	543 (29.6%)	0.266
ASCVD risk score (%)	7.2 ± 7.4	16.4 ± 13.8	6.8 ± 6.8	**<0.001**
Imaging biomarkers				
CAC score (Agatston Units)	0 (0.0–43.4)	129 (13.8–447.8)	0 (0.0–35.4)	**<0.001**
CAC percentile	0 (0.0–63.0)	70.5 (46.5–87.0)	0 (0.0–60.0)	**<0.001**
Number of CAC lesions	0 (0.0–2.0)	4 (1.8–9.0)	0 (0.0–2.0	**<0.001**
EAT volume (cm^3)	77.8 (55.2–104.3)	99.7 (76.5–124.4)	77.2 (54.7–103.3)	**<0.001**
EAT density (HU)	-73.8 ± 4.7	-76.1 ± 4.5	-73.7 ± 4.7	**<0.001**

Results are shown as mean ±SD, *n* (%) or median (inter-quartile range). Event indicates composite endpoint of myocardial infarction or cardiac death. Bold values are statistically significant ($P < 0.05$).
ASCVD, atherosclerotic cardiovascular disease; CAC, coronary artery calcium; EAT, epicardial adipose tissue; HDL, high-density lipoprotein; LDL, low-density lipoprotein.

disease or chest pain, age ≥80 years, pregnancy, prior coronary calcium scanning or significant medical comorbidity such as terminal malignancy, cirrhosis, or end-stage renal disease were excluded. Lipid panel, blood glucose, and blood pressure measurements were completed prior to coronary calcium scoring. Each subject answered a comprehensive questionnaire specific to the EISNER trial to identify cardiovascular risk factors and to describe diet, exercise, and marital status.

2.1.1 Coronary calcium scoring

Non-contrast CT images were acquired for coronary calcium scoring using a multidetector CT or electron beam CT scanner, with 512×512 matrix over a 35 cm field of view, ECG-triggered acquisition, pixel size 0.68×0.68 mm and 2.5- or 3 mm slice thickness. All CT images were reviewed by an experienced cardiac imaging physician. Each scan was analysed using semi-automated software (ScImage, Inc, Los Altos, California) to measure total Agatston coronary calcium score from all calcified lesions with Hounsfield Units > 30 Hounsfield Units (HU).[24]

2.1.2 Prognostic follow-up

Follow-up for subsequent hard adverse cardiovascular events (MI or/and cardiac death) were obtained ∼15 years after the initial visit. Follow-up information was obtained by clinical visits, by detailed questionnaires sent by mail and by telephone contact. Reported events were also verified by the National Death Index query and by comprehensive review of corresponding medical, hospital, and death records by two independent cardiologists blinded to clinical factors and CT data. Three hundred and eighty-four subjects were lost to follow-up and therefore excluded from the analysis.

Our study thus consisted of 1912 subjects (1117 males, 58.4%; 55.8 ± 9.1 years), with a mean follow-up of 14.5 ± 2 years for MI and/or cardiac death. A total of 76 subjects presented an event of MI and/or cardiac death during this follow-up time. Baseline population characteristics are summarized in *Table 1*. The study conformed to the principles outlined in the Declaration of Helsinki and was conducted under the guidelines of the Institutional Review Board. All subjects gave written informed consent for the use of their data.

2.2 Deep learning-based EAT quantification

Deep learning[25] is a sub-category of ML which has demonstrated to outperform state-of-the-art algorithms in many tasks, particularly in computer vision using convolutional neural networks (CNN). We previously demonstrated that CNN can be used for fully automated and robust quantification of EAT volume from non-contrast CAC scoring CT.[26] Our current method is fully automated multitask convolutional neural network which first performs classification to determine if a CT slice is within the heart limits and then performs segmentation of the thorax, heart, and pericardium. This method was validated in a multicentre study of 850 CT scans from multiple scanners, protocols, and sites and performed as well as an expert reader.[27] The deep learning method is incorporated into research software QFAT (version 2.0, Cedars-Sinai Medical Center, Los Angeles, CA) and provides segmentation of EAT, which

can be used to compute the EAT volume as well as the average density in HU.

2.3 Extreme gradient boosting

Extreme gradient boosting (XGBoost)[28] is an optimized implementation of gradient boosting. This ML technique has recently demonstrated high performance in various tasks, such as atrial fibrillation detection from short single-lead electrocardiograms,[29] classification of epileptic patients from language networks in functional magnetic resonance imaging[30] and improved risk stratification in patients with suspected CAD using visually assessed CTA parameters.[31] This method iteratively trains a set of weak learners (or decision trees) using a given set of patient's data or features, to build a combined strong classifier to predict outcomes, such as adverse cardiac events. During the training, more weight was given to patients with an observed event to compensate for unbalanced data.

2.4 Repeated cross-validation and parameters' tuning

To avoid reporting biased results and limit overfitting, ML algorithms are usually validated using cross-validation, separating training, tuning, and testing data. Ten-fold cross-validation is the preferred choice[32] and involves stratifying and dividing the data into 10 folds of equal size. A single fold is used as a test subset, while the other is used for the tuning of the model, and the remaining eight folds are used for training. This process is performed 10 times, always using a different fold for model tuning and testing; stratification is used to ensure a similar distribution of events across the 10 folds. We validated our approach using 10 times repeated 10-fold stratified cross-validation. Repeated 10-fold cross-validation has been shown to reduce the variability due to different initial partitioning (as the process is repeated 10 times using different random partitioning), and has also been shown to have higher performance and lower bias than a bootstrap method.[33] Several configurations of model parameters were assessed to find the optimal set of parameters. For every configuration, tuning and testing ML risk scores were obtained for each subject of the study population. The model configuration providing the best prediction accuracy on the tuning risk scores was selected, and the performance on the associated testing risk scores was reported. The same experiment was repeated for men and women separately to assess for sex-based differences in prognostic importance of covariates.

2.5 Explanation: individualized prediction

ML algorithms are often considered as black boxes as it is difficult to interpret why an algorithm provides accurate predictions for a patient cohort. In this study, we provide a description of individualized prediction made by the algorithm. It allows identification of important patient-specific features and of the role of the feature in the predicted score. To do so, we analyse the specific path a subject takes in the model; in each decision stump (or split) of the model, the individual lands in one of two leaves. Each leaf is associated with a weight: one leaf decreasing the risk of the event happening, the other one increasing the risk. These weights are associated with the variables used to generate the corresponding split. By cumulating all the weights used to refine per-patient prediction for each variable in the model, we can determine whether a parameter has a protective influence or not, depending on the weight sign. By considering the absolute values of cumulated weights, we can also obtain the global contribution of each parameter.

2.6 Statistical analysis

XGBoost as well as statistical analysis was performed in the R software environment (version 3.4.4) on a 64-bit Linux-based personal computer. Continuous variables are reported as mean ± SD or median (inter-quartile range), as appropriate. Categorical variables are reported as frequencies (in per cent) and compared with the χ^2 test. Wilcoxon rank sum test or two-sample *t*-tests were used to compare groups, as appropriate (in *Table 1*). Receiver operating characteristic (ROC) analysis was used to evaluate the prognostic performance of the ML algorithm, atherosclerotic cardiovascular disease (ASCVD) risk score, and CAC score. Comparison between ROC curves was performed by the DeLong test.[34] The highest Youden's index (defined as *J*=sensitivity + specificity-1) was used to define an optimal cut-off and separate patients with low and high ML risk. Log-rank test was then used to assess the difference between Kaplan–Meier curves. Calibration of the ML score was visualized with calibration plots with predicted and actual events. The association with cardiovascular events was also assessed using multivariable Cox regression; the proportional hazards assumption was assessed with Schoenfeld residuals and was valid in all analyses.

3. Results

3.1 Prediction of MI and cardiac death

Both clinical and imaging-based features were important in ML predictions (*Figure 1*). In our patient population, 1049 (55%) out of 1912 subjects had a baseline CAC score of 0. Nevertheless, CAC-related variables were among the most important predictors of MI and cardiac

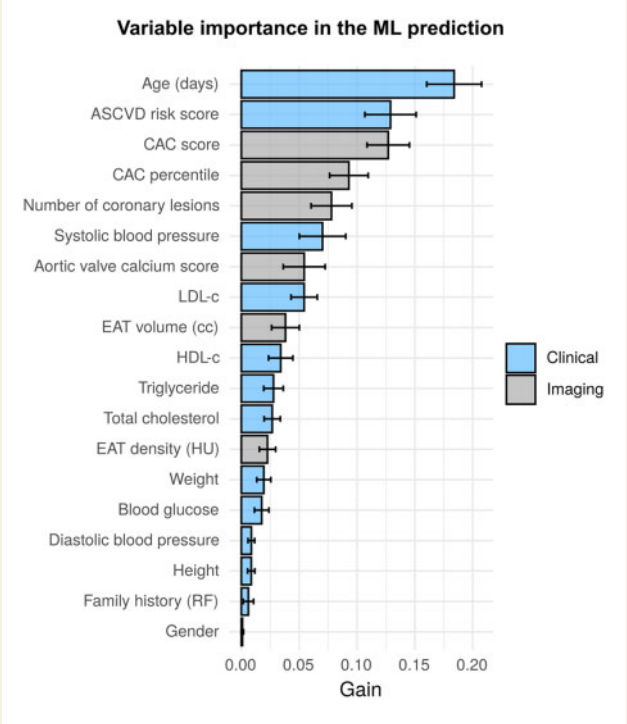


Figure 1 Variable importance for the classification of MI and/or cardiac death (in 1912 subjects). The gain denotes how much a feature contributes to the prediction made by the XGB algorithm. RF, risk factor.

death, and the CAC score had higher variable importance than the ASCVD risk score. Age was the clinical risk factor with the highest variable importance, followed by systolic blood pressure and LDL-cholesterol.

XGB obtained a significantly higher AUC than the ASCVD and the CAC scores (XGB: 0.82; ASCVD: 0.77; CAC: 0.77, $P < 0.05$ for all; *Figure 2*). The ML-score ROC for subjects with CAC < 100 ($N = 1595$) was 0.74 (95% CI: 0.65–0.83) and for subjects with CAC > 400 ($N = 110$) was 0.82 (95% CI: 0.71–0.93). Further, the Integrated Discrimination Improvement for XGB was 8.9% (95% CI: 4.3–13.5) over CAC score and 7.4% (95% CI: 3.3–11.5) over ASCVD risk score.

Figure 3A shows the calibration plot for the ML score, which allows evaluation of the agreement between machine learning scores and the actual distribution of observed events. For comparison, the calibration plots for ASCVD risk scores are also shown (*Figure 3B*); the ASCVD risk score overestimated the risk of events in the higher risk categories (*Figure 2*).

In our data, the ASCVD risk score and imaging measures (CAC, EAT) were the most important prognostic features. We performed ML with and without these features. ML with all measures (ASCVD and imaging measures included) was more predictive for MI and cardiac death than ML with clinical variables only [AUC 0.82 (95% CI: 77–87) vs. 0.78 (95% CI: 0.72–0.84), $P = 0.02$].

3.2 Categorization of ML score

The population was separated into two groups, low and high ML scores, using the maximal Youden's index as an optimal cut-off value (0.0286). At this cut-off, the ML score was associated with a sensitivity and specificity of 0.816 and 0.710, respectively. With the same specificity, CAC and ASCVD risk alone provided a sensitivity of 0.724 and 0.671, respectively. Kaplan–Meier survival curves are shown in *Figure 4*; subjects with a higher ML score were more likely to experience an event (log-rank $P < 0.001$; *Figure 4*).

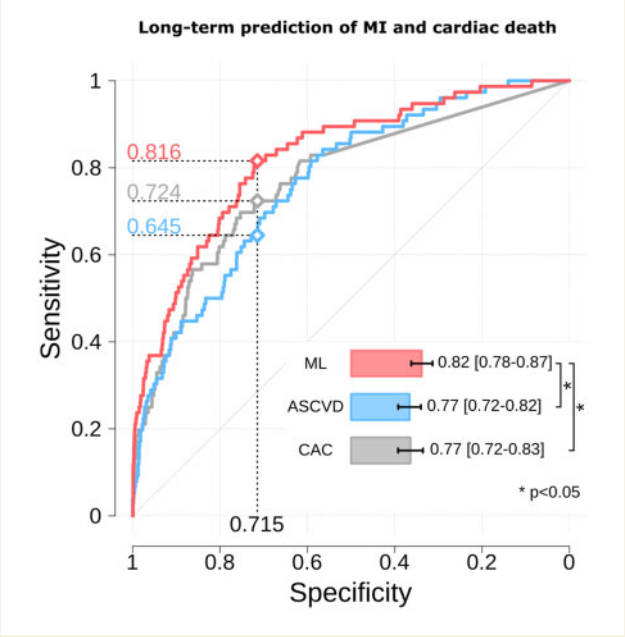


Figure 2 Receiver-operating characteristic curves for the prediction of MI and/or cardiac death (1912 subjects). XGB performed significantly better than the ASCVD risk score ($P < 0.05$) and the CAC scoring ($P < 0.01$). The straight line illustrates the sensitivity, at the same specificity as XGB at maximum Youden's index. At this specificity, the sensitivity was 0.816 for XGB, 0.724 for CAC, and 0.645 for ASCVD risk score.

3.3 Cox regression analysis

In unadjusted analysis, high ML risk was significantly associated with a higher risk of event [unadjusted hazard ratio (HR) 10.38, 95% confidence interval (CI): 5.81–18.53, $P < 0.001$], with a corresponding concordance index of 0.766. When adjusted for CAC and ASCVD risk, EAT volume, gender, and age, the association of high ML risk with events persisted (adjusted HR 2.94, 95% CI: 1.39–6.21, $P = 0.005$), with a concordance index of 0.831. *Table 2* summarizes the multivariable Cox analysis.

3.4 Gender-based analysis

In the sex-specific sub-analysis, age, ASCVD risk score and calcium appeared as important parameters for both men and women (*Figure 5A and B*). However, blood pressure was more important than cholesterol for the prediction in women, while the opposite was observed in men. In both sexes, a high ML risk score was associated with a significantly higher rate of cardiac death or non-fatal MI (*Figure 5C* and *D*).

3.5 Explanation: individualized prediction

To clarify and explain the model prediction for individual patients, we also show in *Figure 6*, two case examples of explaining the ML score prediction: one for a man with no observed event over 14.81 years (*Figure 6A*) and one for a woman with an observed event at 8.79 years after the baseline (*Figure 6B*). The *x* axis corresponds to the ML risk score. The arrows represent the influence of each covariate on the overall prediction; blue and red arrows indicate whether the associated parameters

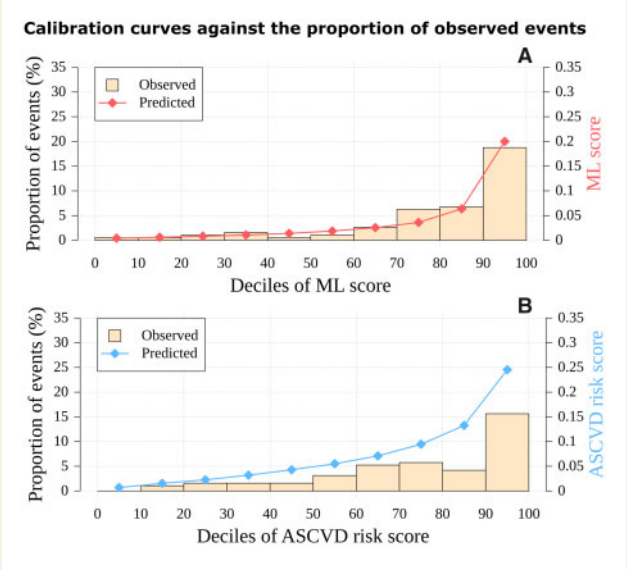


Figure 3 Calibration plots of predicted scores for MI and/or cardiac death prediction ($N = 1912$). (*A*) ML scores in red. (*B*) ASCVD risk scores in blue. The lines and dots represent the average prediction scores per decile. Bars show the average percentage of observed events per decile.

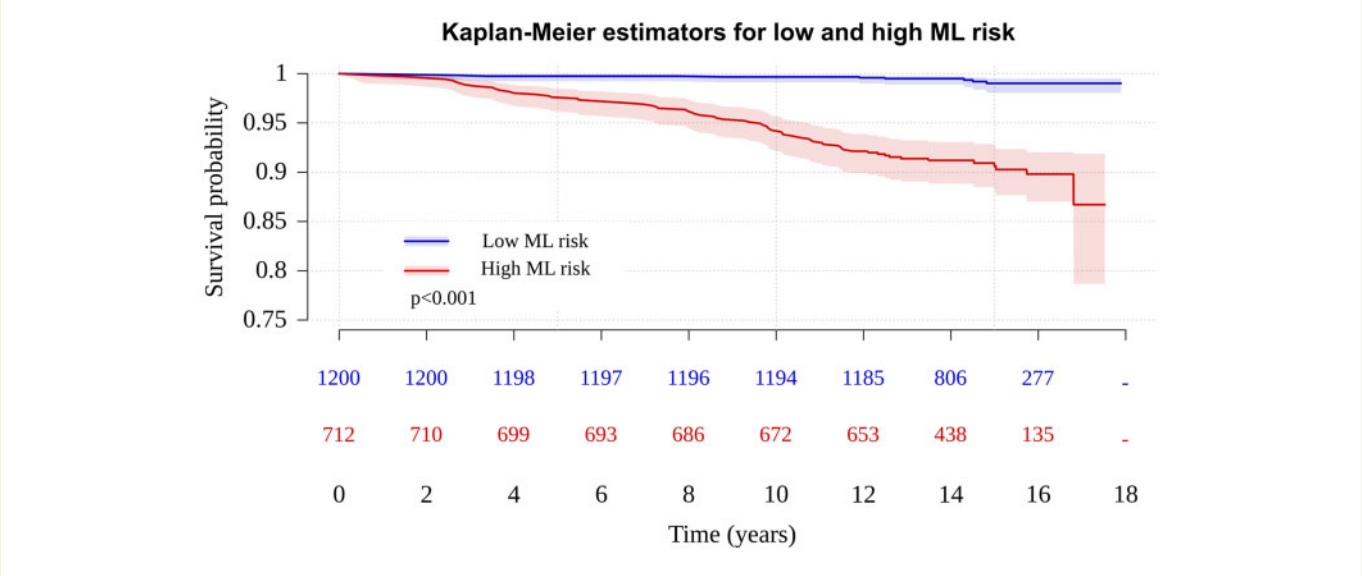


Figure 4 Kaplan–Meier estimator for population ($N = 1912$) with low (blue) and high machine learning risk (red).

Table 2 Multivariable Cox regression for event prediction

	HR (95% CI)	*P*-value
High ML	2.94 (1.39–6.31)	**0.005**
Log2 (CAC score)	1.16 (1.06–1.27)	**0.001**
ASCVD risk score (%)	1.04 (1.01–1.06)	**0.001**
Log2 (EAT volume)	1.43 (0.94–2.16)	0.095
Age (years)	1.01 (0.98–1.04)	0.670
Gender (male)	0.96 (0.58–1.60)	0.885

Event indicates composite endpoint of myocardial infarction or cardiac death. Bold values are statistically significant ($P < 0.05$).
ASCVD, atherosclerotic cardiovascular disease; CAC, coronary artery calcium (AU); CI, confidence interval; EAT, epicardial adipose tissue (cm^3); HR, hazard ratio; ML, machine learning.

decrease (blue) or increase (red) the risk of future events. The combination of all covariates' influence provides the final ML risk score. The red and blue colours provide the separation between low and high ML risk.

4. Discussion

In this prospective trial, ML demonstrated high performance in risk assessment for MI and cardiac death in asymptomatic subjects. By objectively combining clinical data and quantitative CT measures, ML provided significantly superior risk prediction compared with the ASCVD risk score or CAC score. These promising results suggest that ML has a potential for clinical implementation to improve risk assessment.

In our study, objective integration of the standard clinical variables with calcium and EAT quantification was performed during the optimization of the ML model. The model takes full advantage of the CAC and EAT volume biomarkers and combines them with other metrics of cardiovascular risk. Such ML integration, from clinical data to image-based quantitative parameters, can thus improve cardiovascular risk stratification for the patient, using all relevant covariates rather than individual measures. Our approach was also validated with repeated 10-fold cross-validation to provide a robust estimation of prediction accuracy with minimal bias. It significantly outperformed the standard ASCVD risk score, as well as the CAC score, well-known for its prognostic importance. To our knowledge, our study is first to show that an objective machine learning score integrating clinical data and *quantitative* CT assessment, after CAC scoring, can significantly improve risk assessment and provide personalized patient-specific explanations of the prediction.

In our study, variable importance showed that both clinical and imaging-based parameters were important to provide an optimal risk estimation. As expected, age, ASCVD risk score and calcium-related information were the highest contributor to ML prediction, but other parameters were also important to provide more accurate estimation. For example, systolic blood pressure, even though it is already included in the ASCVD risk score, still appeared as an independent strong covariate for the risk prediction. LDL cholesterol, another strong contributor to the overall ML predictions, was the most important lipid measurement. EAT volume also showed a contribution to the results, more than HDL cholesterol or triglycerides. Notably, our rank of variable importance (*Figure 1*) corresponds, in general terms, to the observed differences in clinical variables among subjects who did and did not have cardiovascular events in our study. For example, subjects who had events at a greater age and had substantially higher ASCVD scores, CAC abnormality, and SBP (*Table 1*). These were all parameters which had high variable importance by ML. Risk factors of smoking, and diabetes are included in the ASCVD risk score, which is in turn higher when these risk factors are present. The ASCVD risk score was therefore specifically included in the ML model, to extrapolate and generalize to external populations with higher prevalence of these key risk factors.

Our study extends several other recent studies which report the use of ML for risk assessment and prediction of cardiovascular events with cardiac CT.[23,35,36] These previous studies were based on the MESA population and the CONFIRM registry, for asymptomatic subjects and patients with suspected CAD, respectively. Various approaches of ML

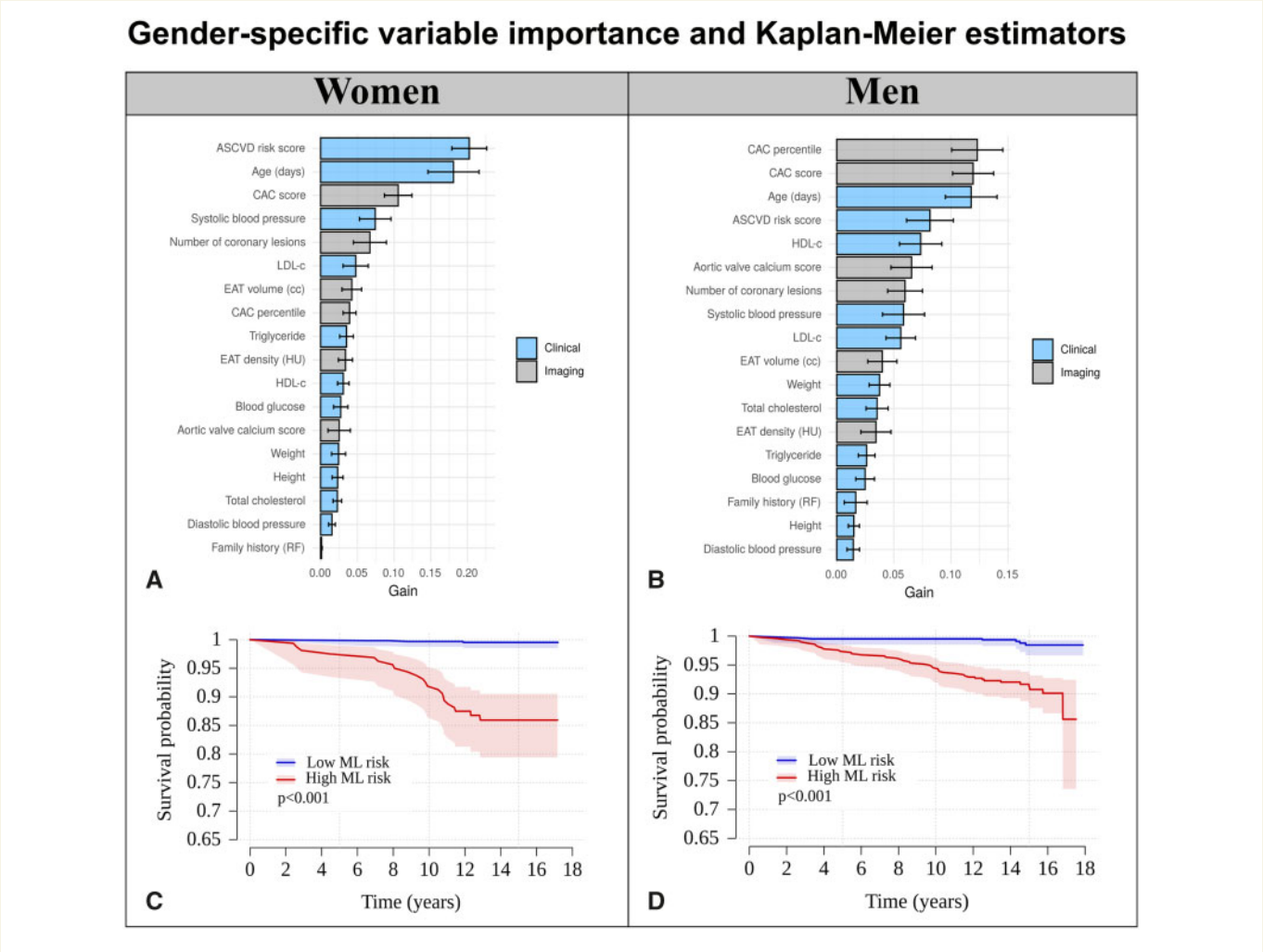


Figure 5 Variable importance in ML classification for women (*A*, *N* = 795) and men (*B*, *N* = 1117). Kaplan–Meier curves for subjects with high (red) and low ML risk (blue) in women (*C*) and (*D*).

were considered, such as Random Forests, support vector machines or LogitBoost, outperforming standard clinical metrics in prediction accuracy. Standard risk factors and clinical parameters were used alone as predictors or combined with image-based quantitative variables, providing richer information to enhance risk prediction. The use of manual measurement and visual assessment for the extraction of image features, however, adds burden to the clinical routine procedure and induces a bias due to inter-expert variability. In our approach, EAT quantifications were performed automatically from the calcium scoring scan and integrated with other predictors to perform risk assessment, without increasing the physician's workload or requiring an additional exam for the patient. Additionally, while more classical ML methods were used in the previous studies, in our study, we investigated the recent XGBoost algorithm, which has shown to outperform other algorithms for sets of features in a variety of tasks. Another advantage of using this model is the possibility to explicitly describe the influence of each variable for an individualized prediction. Such prediction breakdown on a subject basis has a real interest in clinical practice by personalizing prevention and potentially driving and reinforcing therapeutic strategies.

With constant growth of artificial intelligence across various disciplines, especially in cardiology,[37] robust cardiac risk assessment will benefit from quantification of automated imaging biomarkers, increasing the relevant information available for clinical decision making. Deep learning can, for example, perform fully automated calcium scoring using CNN[38,39] as well as epicardial adipose tissue measurement from non-contrast cardiac CT and quantify coronary artery stenosis from coronary CT Angiography;[40,41] such imaging biomarkers could be automatically included in event prediction without requiring tedious and repetitive manual expert measurements.

In this study, we have been only able to discuss the performance of machine learning in a *post hoc* analysis of a clinical trial with long-term follow-up; but we were not able to assess its prospective practical implementation. In the future, we foresee machine learning working in the background of standard coronary calcium scoring and electronic reporting software, gathering the variables automatically and allowing 'on-the-fly' risk score computation by integrating all relevant measures of clinical risk and imaging biomarkers from coronary calcium scoring scans. We expect that such personalized prevention tools would help physicians

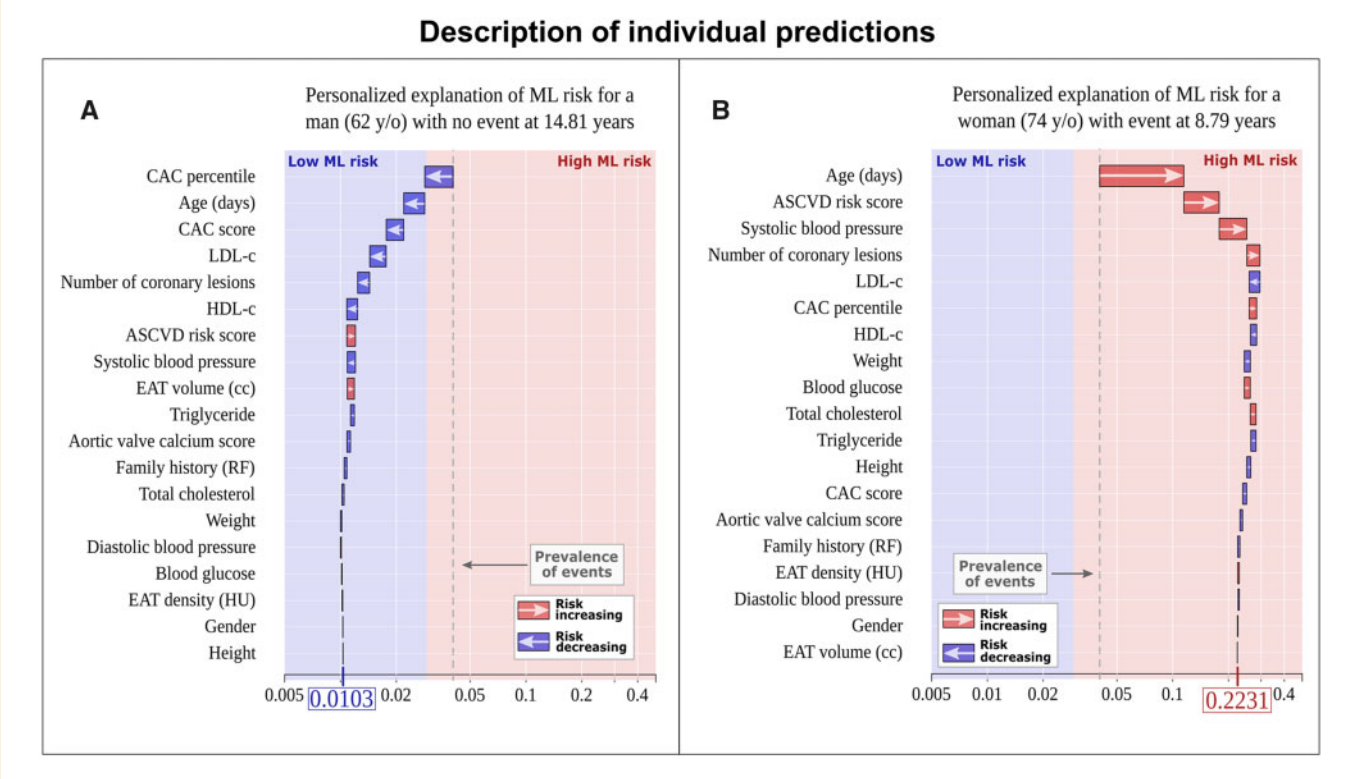


Figure 6 Explanations of individual prediction with subject-specific feature importance for a 62-year-old man with no event (*A*) and a 74-year-old woman with an event at 8.79 years (*B*). The *x* axis corresponds to the ML risk score. The arrows represent the influence of each covariate on the overall prediction; blue and red arrows indicate whether the covariates decrease (blue) or increase (red) the risk of future events. The combination of all covariates' influence provides the final ML risk score. For the male subject in (*A*), there is a low ML risk score (0.0103), with an ASCVD risk score of 7.25%, CAC score of 0 and 0th CAC percentile. For the female subject in (*B*), there is a high ML risk score (0.2231), with an ASCVD risk score of 30.36%, a CAC score of 94 and 63rd CAC percentile. The red and blue colours provide the separation between low and high ML risk by optimal Youden's index and grey dashed lines correspond to the base risk obtained from the rate of event in the population (~4%).

find the right answers for their patients,[42] whose 'lives and medical histories shape the algorithms'.[43]

4.1 Limitations

In addition to its single-centre design, our study had several limitations. First, the low number of subjects in the trial and the small rate of observed events (<4%) over a long-term follow-up may have led to overfitting. While we tried to address this by requiring more regularization during model training, a larger number of subjects would increase in the model's capacity and likely lead to a more accurate risk estimation. Our study lacks external validation by an independent cohort; such validation would require an asymptomatic patient population with lipid panel and coronary and aortic calcium with long-term follow-up as well as image data, for measurement of new imaging biomarkers of epicardial adipose tissue volume and density. We have also not compared our results with those obtained with a well-validated risk calculator derived from the MESA population[44] but rather focused on machine learning integration of all available data and compared the result with the widely utilized ASCVD risk score. Moreover, the EISNER trial included only asymptomatic subjects with no prior history of CAD or significant co-morbidity. Our results are therefore representative of the asymptomatic population undergoing coronary calcium scoring for cardiovascular risk assessment and may need re-evaluation in symptomatic subjects undergoing coronary calcium scoring or coronary CT Angiography.

5. Conclusion

In this prospective study, machine learning used to integrate clinical and quantitative imaging-based variables improved prediction of MI and cardiac death compared with standard clinical risk assessment and provided explicit explanation of individualized risk prediction. Following further validation, such a personalized paradigm could potentially be used to improve cardiovascular risk assessment.

Authors' contributions

F.C., P.J.S., S.C., D.D.

- Substantial contributions to the design of the work; and acquisition, analysis and interpretation of data; AND
- Drafting of the work and revising it critically for intellectual content; AND
- Final approval of the version to the published; AND

- Agreement to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

M.G., A.R., A.R., B.K.T.

- Substantial contributions to the acquisition, analysis, or interpretation of data for the work; AND
- Drafting the work or revising it critically for important intellectual content; AND
- Final approval of the version to be published; AND
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- Final approval of the version to be published; AND
- Agreement to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

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D.D.—Guarantor of the overall integrity of the article.

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Translational Perspective

We aimed to evaluate the performance of machine learning integrating clinical risk and imaging biomarkers (including coronary calcium score and epicardial adipose tissue measures from non-contrast CT) to predict long-term risk of myocardial infarction and cardiac death in asymptomatic subjects. Our results from the EISNER trial showed that machine learning integrating clinical risk and imaging biomarkers outperforms standard clinical risk assessment for the prediction of myocardial infarction and cardiac death over long-term follow-up (with area-under-receiver-operating-curve of 0.82) and provides explicit explanation of individualized risk prediction. Such personalized integrated prevention tools could help physicians find the right answers for their patients.